

Beyazit Yalcinkaya

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Research Interests

Formal methods, AI/ML, AI safety, reinforcement learning, multi-task reinforcement learning, multi-agent reinforcement learning, representation learning, human-computer interaction.

Education

2021 - Present	Ph.D. in Computer Science Advised by Sanjit A. Seshia <i>University of California, Berkeley, USA</i>
2015 - 2020	B.Sc. in Computer Engineering Valedictorian of the department <i>Middle East Technical University, Ankara, Turkey</i>

Research Experience

2021 - Present	Graduate Student Researcher Supervised by Sanjit A. Seshia <i>University of California, Berkeley, USA</i>
Summer 2023	Applied Scientist Intern Supervised by Ankush Desai <i>Amazon Web Services, East Palo Alto, USA</i>
Summer 2022	Applied Scientist Intern Supervised by Ankush Desai <i>Amazon Web Services, Cupertino, USA</i>
2017 - 2020	Undergraduate Research Assistant Supervised by Ebru Aydin Gol <i>University of California, Berkeley, USA</i>
Summer 2019	Research Intern Supervised by George Candea <i>École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland</i>
Summer 2018	Research Intern Supervised by Björn B. Brandenburg <i>Max Planck Institute for Software Systems, Kaiserslautern, Germany</i>

Teaching Experience

Spring 2026	Graduate Student Instructor
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	CS 189/289A Introduction to Machine Learning University of California, Berkeley, USA
Fall 2022	Graduate Student Instructor EE/CS 149/C249A Introduction to Embedded Systems University of California, Berkeley, USA
Fall 2017	Undergraduate Teaching Assistant CENG 230 Introduction to C Programming Middle East Technical University, Ankara, Turkey

Refereed Conference and Journal Papers

- [1] *Beyazit Yalcinkaya, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, **Learning Task-Conditioned Policies with Natural Language and Graph Embeddings of Formal Specifications**, under review.*
- [2] *Beyazit Yalcinkaya, Marcell Vazquez-Chanlatte, Ameesh Shah, Hanna Krasowski, and Sanjit A. Seshia, **Automata-Conditioned Cooperative Multi-Agent Reinforcement Learning**, under review.*
- [3] *Beyazit Yalcinkaya, Niklas Lauffer, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, **Provably Correct Automata Embeddings for Optimal Automata-Conditioned Reinforcement Learning**, 2nd International Conference on Neuro-symbolic Systems (NeuS), 2025.*
- [4] *Beyazit Yalcinkaya*, Niklas Lauffer*, Marcell Vazquez-Chanlatte*, and Sanjit A. Seshia, **Compositional Automata Embeddings for Goal-Conditioned Reinforcement Learning**, NeurIPS, 2024. *Equal contribution.*
- [5] *Beyazit Yalcinkaya, Hazem Torfah, Daniel J. Fremont, and Sanjit A. Seshia, **Compositional Simulation-Based Analysis of AI-Based Autonomous Systems for Markovian Specifications**, International Conference on Runtime Verification (RV), 2023.*
- [6] *Beyazit Yalcinkaya, Hazem Torfah, Ankush Desai, and Sanjit A. Seshia, **ULGEN: A Runtime Assurance Framework for Programming Safe Cyber-Physical Systems**, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2023.*
- [7] *Niklas Lauffer*, Beyazit Yalcinkaya*, Marcell Vazquez-Chanlatte, Ameesh Shah, and Sanjit A. Seshia, **Learning deterministic finite automata decompositions from examples and demonstrations**, Formal Methods in Computer-Aided Design (FMCAD), 2022. *Equal contribution.*
- [8] *Mert Ergurtuna, Beyazit Yalcinkaya, and Ebru Aydin Gol, **An automated system repair framework with signal temporal logic**, Acta Informatica, 2022.*
- [9] *Beyazit Yalcinkaya and Ebru Aydin Gol, **Clock reduction in timed automata while preserving design parameters**, International Conference on Formal Methods in Software Engineering (FormalISE), 2019.*
- [10] *Beyazit Yalcinkaya, Mitra Nasri, and Björn B. Brandenburg, **An exact schedulability test for non-preemptive self-suspending real-time tasks**, Design, Automation and Test in Europe Conference (DATE), 2019.*

Awards and Honors

Fall 2021	University of California, Berkeley, EECS Department Fellowship
2018 - 2020	Scientific and Technological Research Council of Türkiye Research Fellowship
2019	Summer@EPFL Fellowship